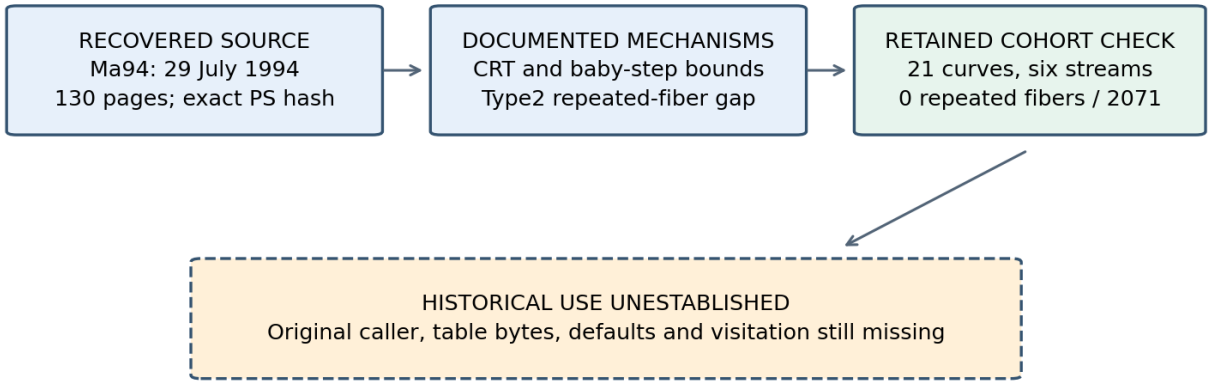
The 1994 implementation thesis

Actual resource controls, a documented exceptional case, and the remaining caller gap

27 October 2026 | T-5518 | implementation-sources-01

Historical selector unresolved. This is a primary-source review and crosswalk to completed evidence. There are zero new scientific trials. The goal remains the actual mechanism accounting for all eleven reconstructed omissions while retaining ten publications across six streams.

New source recovered: Markus Maurer's 130-page Saarland diploma thesis has a title-page date of 29 July 1994. The original gzip was recovered from an archived university FTP mirror; its PostScript header is dated17 October 1994. A 2006 capture date establishes preservation then, not the original 1994 custody.



T-5518 implementation-sources-01 | Source interpretation of existing evidence; no new scientific trials

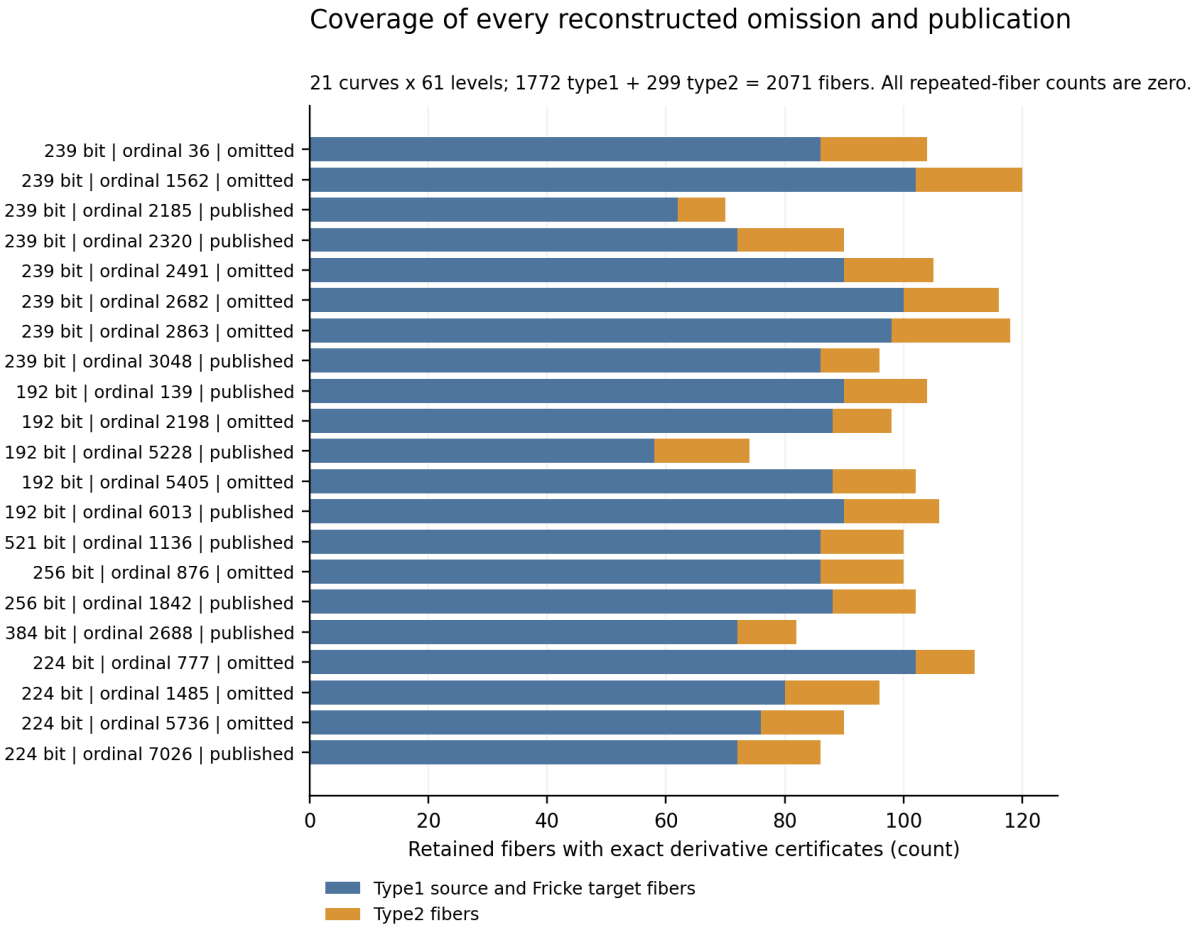
The thesis identifies the C/C++ program **eco**, source-file/function names, relation-file formats, configurable resource controls and command-line modes. It does not identify a NIST/ANSI invocation or a candidate-decision log.

The exact public gzip is 281,859 bytes, SHA256:
`a 0bfcc09e9d58bfefac531bf9a9037f42b317c36f7e5681f273b283cf6b1b3e3`

The author-hosted ANTS 1994 counting paper and the 27 April 1995 Buchmann-Muller-Shoup report were also obtained and read. The former has a 2001 PDF conversion date; no byte identity to the proceedings edition is claimed. Source bytes and renderings remain private under .work/.

Every omission and publication covered

The retained modular-coordinate audit covers 21 curves, 61 levels 3..293, 1185 rational modular parameters and 2071 fibers. The new script checks every p,j and full p,b,n,UID binding before summarizing those existing observations.



Existing modular-coordinate-guards evidence, later LiDIA tables, ell=3..293. Counts are exact; historical table identity is unknown.

Ma94 p 93: type 2 neighbor reconstruction is unspecified when G_l(A(tau),Y) has repeated roots. In the retained later-table audit all 299 type 2 fibers, and all 1772 type 1 source/Fricke fibers, have coprime derivative gcds. The condition occurs on none of the 21 curves. An existing level 53 positive control over F1009 does trigger it.

These are exact retained counts, without a statistical uncertainty model. They are not a fresh run of eco 1994. Original 1994 coefficient bytes and the 1997 caller remain unauthenticated. Full fields, equations, orders, seeds, ordinals and UIDs are retained in cohort-crosswalk.csv.

What the implementation actually controls

pp 65-66: baby-step memory
A configurable K_B,max shifts moduli to giant steps and rebalances them. It bounds a table, not the eigenvalue list at a particular prime. No fixed 32 or 2048 acceptance rule is supplied.

p 111: total combinations
MAXKOMB concerns the combined CRT candidate population. Its numerical value is not supplied. It is distinct from an eigenvalue-list length and from the minimum degree of a division-polynomial factor.

pp 117-118: caller modes
-moduls supplies a prime list; -noadeg suppresses Atkin-degree work only with that list; -rsplit skips primes missing saved split data; -proof requests order proof. Actual invocation, defaults and files remain unknown.

pp 83-93: neighbor formulas
Given coefficients supply E4=-a/3 and E6=-b/2. The displayed interfaces request no cube root of j. Type 1 DJ=0 has a square-root fallback; type 2 repeated roots have the explicit gap checked on page 2.

pp 94-100: eigenvalue matching
Signed positive-half candidates and a y-only heuristic are described. The previous independent 411-kernel comparison has no wrong first-y match on any cohort kernel; it includes 228 omission and 183 publication kernels over 22 represented levels.

pp 39-46: CM shortcuts
Norm-based special counters cover supersingular curves and base-field isogeny classes containing j=0/1728. The previous all 21 comparison finds no cohort member on those routes, with generic-j positive controls.

Literal pseudocode caveat. The image on printed p 111 really initializes KOMB=1 and prints an AND guard requiring KOMB>MAXKOMB. For every positive integer MAXKOMB this is initially false for every curve. A literal copy would visit no odd auxiliary prime. This is a documentary inconsistency, not an observed C-program defect. An OR repair would be a reconstruction choice.

Provenance and the next discriminating evidence

Primary passages read. Ma94: title/date; pp 22-28, 39-46, 59-61, 65-67, 83-100, 110-114, 117-126. Title and PDF pages 67, 94, 96, 112, 119 were inspected visually. Printed page = PDF page minus one. The two short reports were read completely; their date and key algorithm pages were inspected.

Retained imports. CROSSWALK.json pins the prior cohort, fiber audit and its bundle, complete y-match audit and bundle, and independent CM-route result and bundle. build_crosswalk.py --check reconstructs the complete CSV and metadata from those records. It neither invokes a point counter nor changes a frozen result.

Access limits. The archived project page and report directory now establish both thesis filenames. Exact Le94 archive requests return an empty array; its contents remain unobtained. Broad 100-row CDX outputs are bounded. Two wildcard queries returned general-domain rows and are unused as filename evidence. Failed CLI and catalog attempts remain in the retrieval receipts. No copy request or outgoing contact was sent.

What the evidence leaves open. The P-224 cube/splitting pattern and the retrospective ell 199 separator remain leads. This source supplies neither a cube-root-of-j prerequisite nor the two fitted caps. Original function bodies, coefficient tables, defaults, invocation, root order, error policy and candidate visitation remain unknown. No endomorphism, conductor, transfer or cryptographic weakness conclusion follows from these passages.

Next sources. Recover the named C bodies, especially buildLlist, choose_relations, compute_klist, binsearch and verify_order; recover Le94 and authenticated coefficient data; obtain the original Reiter/NIST/ANSI caller or run log. Freeze any newly recovered rule before comparing every omission and publication without retuning. Reconstructed omissions remain distinct from documented rejection events.

Primary links
Ma94 archive: web.archive.org/web/20060216012606id_/http://www.informatik.tu-darmstadt.de:80/ftp/pub/TI/reports/maurer.diplom.ps.gz
ANTS author copy: shoup.net/papers/counting.pdf
1995 report: publikationen.sulb.uni-saarland.de/handle/20.500.11880/25886
Full Markdown report, exact receipts, editable SVGs and 21-curve CSV accompany this digest.

Canonical parent diagrams were checked and their mathematical statements remain unchanged. No formal hypothesis status, isogeny edge or historical-selector attribution is added. The full goal remains active.